Random walks

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Definition

Let $\{Y_k \colon k \geqslant 1\}$ be a sequence of independent and identically distributed random variables, such that $\mathbb{P}(Y_k = 1) = p$ and $\mathbb{P}(Y_k = -1) = q \equiv 1 - p$. Let

$$X_n = \begin{cases} 0, & n = 0 \\ Y_1 + Y_2 + \cdots + Y_n, & n \geqslant 1 \end{cases}$$

The sequence $\{X_n \colon n \geqslant 1\}$ is a discrete-time stochastic process known as a random walk with parameter p.

- The state space of the process is $\mathbb{Z}$.
- If $p = q = 1/2$, we say that the random walk is symmetric.

Random walk

A trajectory of this process can be visualised as a particle moving randomly in one dimension, starting from the origin.

If the particle is at coordinate k at time n, then at time $n+1$, it will either move

- *to coordinate $k+1$, with probability p,*

$$\mathbb{P}(X_{n+1} = k+1 \,|\, X_n = k) = p$$

- *or to coordinate $k-1$, with probability q,*

$$\mathbb{P}(X_{n+1} = k-1 \,|\, X_n = k) = q$$

Random walk

An alternative interpretation considers a game involving two players, A and B, who repeatedly toss a coin that lands heads with probability p.

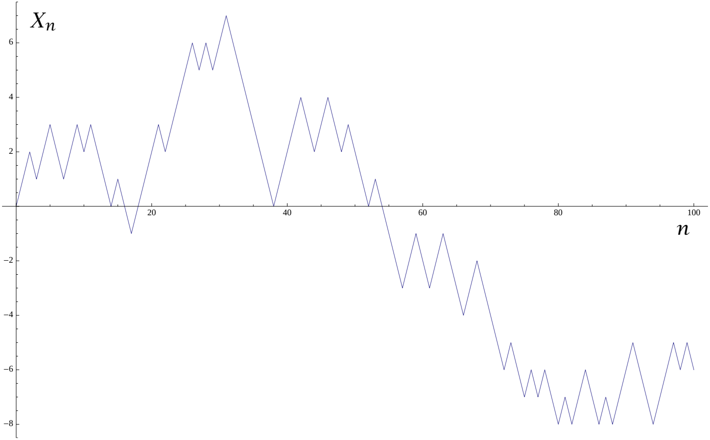
Let H_n and T_n be the number of heads and tails in the first n tosses, and consider

$$X_n = H_n - T_n, \quad n \geqslant 1$$

The state of the game at time n (after n tosses) can be described as follows:

- *If $X_n > 0$, we say that player A is leading;*
- *If $X_n < 0$, we say that player B is leading;*
- *If $X_n = 0$, the game is in a draw.*

Random walk



First-order probability function

Note that $X_n = k$ if and only if, in the first n tosses, the number of heads r and the number of tails s satisfy $k = r - s$. Given that $s + r = n$, it follows that $k = 2r - n$, and thus $r = (n + k)/2$, where $r = 0, 1, \ldots, n$.

Proposition

At time n, the particle is at coordinate k with probability

$$\mathbb{P}(X_n = k) = \binom{n}{\frac{n+k}{2}} p^{\frac{n+k}{2}} q^{\frac{n-k}{2}},$$

where $k = -n, -n + 2, \ldots, n - 2, n$. In particular, if the random walk is symmetric,

$$\mathbb{P}(X_n = k) = \binom{n}{\frac{n+k}{2}} 2^{-n}.$$

Mean and variance

Since $X_n = H_n - T_n = 2H_n - n$, we have

$$\mathbb{E}(X_n) = \mathbb{E}(H_n) - \mathbb{E}(T_n) = n(p-q)$$

$$\mathrm{Var}(X_n) = 4\,\mathrm{Var}(H_n) = 4npq$$

Furthermore, by the Central Limit Theorem:

$$\frac{X_n - n(p-q)}{\sqrt{4npq}} \xrightarrow{d} \mathrm{N}(0,1)$$

If the random walk is symmetric, $\mathbb{E}(X_n) = 0$, $\mathrm{Var}(X_n) = n$, and

$$\frac{X_n}{\sqrt{n}} \xrightarrow{d} \mathrm{N}(0,1)$$

Spatial and temporal homogeneity

Let us assume that the random walk starts from the origin, that is, let $X_0 = 0$. If $m > n$, then

$$\mathbb{P}\left(X_m = b \,|\, X_n = a\right) = \mathbb{P}\left(X_{m-n} = b - a\right),$$

because each term is equal to $\mathbb{P}\left(\sum_{k=1}^{m-n} Y_k = b - a\right)$.

Proposition

- *(Spatial homogeneity) For all* $r > 0$,

$$\mathbb{P}(X_m = b \,|\, X_n = a) = \mathbb{P}(X_m = b + r \,|\, X_n = a + r)$$

- *(Temporal homogeneity) For all* $p > 0$,

$$\mathbb{P}(X_m = b \,|\, X_n = a) = \mathbb{P}(X_{m+p} = b \,|\, X_{n+p} = a)$$

Independent increments

Let $\mathcal{S}_n = \{-n, -n+2, \ldots, n-2, n\}$ be the set of positions that the random walk can visit at time n.

Let s represent the increment $X_m - X_n$ between times n and m, where $n < m$.

Therefore,

$$\begin{aligned}
\mathbb{P}(X_m - X_n = s) &= \sum_{u \in \mathcal{S}_n} \mathbb{P}(X_m = u + s \,|\, X_n = u)\, \mathbb{P}(X_n = u) \\
&= \mathbb{P}(X_{m-n} = s) \sum_{u \in \mathcal{S}_n} \mathbb{P}(X_n = u) \\
&= \mathbb{P}(X_{m-n} = s)
\end{aligned}$$

Independent increments

More generally, if $n < m < r$, then

$$\mathbb{P}\left(X_r = c \,|\, X_m = b, X_n = a\right) = \mathbb{P}\left(X_r = c \,|\, X_m = b\right)$$

because each term is equal to $\mathbb{P}\left(\sum_{i=1}^{r-m} Y_i = c - b\right)$.

This is basically the Markov property: conditional on the present, the future is independent of the past.

Therefore,

$$\begin{aligned}\mathbb{P}\left(X_r = c \,|\, X_m = b, X_n = a\right) &= \mathbb{P}\left(X_r = c \,|\, X_m = b\right) \\ &= \mathbb{P}(X_{r-m} = c - b) = \mathbb{P}(X_r - X_m = c - b),\end{aligned}$$

Independent increments

In summary, we conclude that the random walk has independent increments.

$$
\begin{aligned}
\mathbb{P}(X_n = a, X_m - X_n = b - a, X_r - X_m = c - b) \\
&= \mathbb{P}\left(X_n = a, X_m = b, X_r = c\right) \\
&= \mathbb{P}\left(X_r = c \,|\, X_m = b, X_n = a\right) \mathbb{P}\left(X_m = b, X_n = a\right) \\
&= \mathbb{P}\left(X_r = c \,|\, X_m = b\right) \, \mathbb{P}\left(X_m = b \,|\, X_n = a\right) \, \mathbb{P}(X_n = a) \\
&= \mathbb{P}(X_n = a)\, \mathbb{P}(X_{m-n} = b - a)\, \mathbb{P}(X_{r-m} = c - b) \\
&= \mathbb{P}(X_n = a)\, \mathbb{P}(X_m - X_n = b - a)\, \mathbb{P}(X_r - X_m = c - b)
\end{aligned}
$$

Independent increments

These results can be straightforwardly generalised to $l \geqslant 2$ instants of time.

Proposition

Let $0 < n_1 < n_2 < \cdots < n_l$. *We have:*

$$\begin{aligned}
&\mathbb{P}(X_{n_1} = a_1, X_{n_2} = a_2, \ldots, X_{n_l} = a_l) \\
&\quad = \mathbb{P}(X_{n_1} - X_0 = a_1, X_{n_2} - X_{n_1} = a_2 - a_1, \cdots \\
&\qquad\qquad \cdots X_{n_l} - X_{n_{l-1}} = a_l - a_{l-1}) \\
&\quad = \mathbb{P}(X_{n_1} - X_0 = a_1)\,\mathbb{P}(X_{n_2} - X_{n_1} = a_2 - a_1)\cdots \\
&\qquad\qquad \cdots \mathbb{P}(X_{n_l} - X_{n_{l-1}} = a_l - a_{l-1}) \\
&\quad = \mathbb{P}(X_{n_1} = a_1)\,\mathbb{P}(X_{n_2 - n_1} = a_2 - a_1)\cdots\mathbb{P}(X_{n_l - n_{l-1}} = a_l - a_{l-1})
\end{aligned}$$

First visit to a negative coordinate

Let T denote the random time at which the particle first reaches a negative coordinate.

$$T = \min\{n \geqslant 1 \colon X_n = -1\}$$

In the context of the coin-tossing game, T represents the first instance at which player B takes the lead.

Let us calculate $\mathbb{P}(T = n)$.

Notice that $\mathbb{P}(T = 2k) = 0$ for all $k \geqslant 1$. because if $X_n = -1$, then n must be and odd integer.

First visit to a negative coordinate

For instance,

$$\begin{aligned}\mathbb{P}(T=1) &= q\\ \mathbb{P}(T=2) &= 0\\ \mathbb{P}(T=3) &= pq^2\\ \mathbb{P}(T=4) &= 0\\ \mathbb{P}(T=5) &= 2p^2q^3\end{aligned}$$

- We will see that if $p > q$, the particle may never visit a negative coordinate. In this case,

$$\mathbb{P}(T=\infty) \equiv 1 - \sum_{n\geqslant 1} \mathbb{P}(T=n) > 0,$$

and we say that the random variable T is defective.

First visit to a negative coordinate

For any odd integer $n \geqslant 3$, we have the following relationship:

$$P(T = n) = \mathbb{P}(T = n, X_1 = 1) = p\,\mathbb{P}(T = n \mid X_1 = 1)$$

Define $Z = \min\{m \geqslant 2 \colon X_m = 0\}$ as the time of the first return to zero.

Therefore, by the TPT, we have

$$\begin{aligned}
&\mathbb{P}(T = n \mid X_1 = 1) \\
&\qquad = \sum_{m=2}^{n-1} \mathbb{P}(T = n \mid Z = m, X_1 = 1)\,\mathbb{P}(Z = m \mid X_1 = 1)
\end{aligned}$$

First visit to a negative coordinate

Furthermore, notice that

- $\mathbb{P}(Z = m \,|\, X_1 = 1) = \mathbb{P}(T = m - 1)$
- $\mathbb{P}(T = n \,|\, Z = m, X_1 = 1) = \mathbb{P}(T = n - m)$

Therefore, if we denote the probability $\mathbb{P}(T = n)$ as r_n, we have the following recurrence equation:

For all odd integer $n \geqslant 3$,

$$r_n = p \sum_{m=2}^{n-1} r_{m-1} r_{n-m} = p\,(r_1 r_{n-2} + r_3 r_{n-4} + \cdots + r_{n-2} r_1),$$

where $r_1 = q$.

First visit to a negative coordinate

For instance,

$$
\begin{aligned}
P(T=1) &= r_1 = q \\
\mathbb{P}(T=3) &= r_3 = p\, r_1^2 = pq^2 \\
\mathbb{P}(T=5) &= r_5 = p\,(r_1 r_3 + r_3 r_1) = 2p^2 q^3 \\
\mathbb{P}(T=7) &= r_7 = p\,(r_1 r_5 + r_3^2 + r_5 r_1) = 5p^3 q^4 \\
\mathbb{P}(T=9) &= r_9 = p\,(r_1 r_7 + r_3 r_5 + r_5 r_3 + r_7 r_1) = 14p^4 q^5 \\
&\cdots\cdots\cdots\cdots\cdots
\end{aligned}
$$

First visit to a negative coordinate

Let $G_T(s) = \sum_{n \geqslant 0} r_n \, s^n$ be the generating function of the sequence $\{r_n \colon n \geqslant 0\}$. (We take $r_{2k} = 0$ for all $k \geqslant 0$.)

Notice that

$$
\begin{aligned}
& p\, s\, \big(G_T(s)\big)^2 \\
&\qquad = ps\,(r_1 s + r_3 s^3 + r_5 s^5 + \cdots)(r_1 s + r_3 s^3 + r_5 s^5 + \cdots) \\
&\qquad = p\, r_1^2 s^3 + p(r_1 r_3 + r_3 r_1) s^5 + p\,(r_1 r_5 + r_3^2 + r_5 r_1) s^7 + \cdots \\
&\qquad \cdots + p\,\underbrace{(r_1 r_{n-2} + r_3 r_{n-4} + \cdots + r_{n-2} r_1)}_{r_n}\, s^n + \cdots \\
&\qquad = r_3 s^3 + r_5 s^5 + r_7 s^7 + \cdots + r_n s^n + \cdots = G_T(s) - q\, s
\end{aligned}
$$

First visit to a negative coordinate

Hence, the generating function $G_T(s)$ satisfies the quadratic equation

$$p\,s\,(G_T(s))^2 - G_T(s) + qs = 0$$

Solving for $G_T(s)$ we deduce that

$$G_T(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps} = \frac{2qs}{1 + \sqrt{1 - 4pqs^2}}$$

- We choose the sign of the square root to guarantee that $G_T(s)$ remains finite at $s = 0$.
- $G_T(s)$ is well-defined if $1 - 4pqs^2 \geqslant 0$, which holds when $s \in [-R, R]$, where $R = 1/(2\sqrt{pq})$. Therefore, if $p = q = 1/2$, then $R = 1$. In all other cases, $R > 1$.

First visit to a negative coordinate

For a symmetric random walk ($p = q = 1/2$):

$$G_T(s) = \frac{s}{1 + \sqrt{1 - s^2}}$$

$$= \frac{1}{2}s + \frac{1}{8}s^3 + \frac{1}{16}s^5 + \frac{5}{128}s^7 + \frac{7}{256}s^9 + \frac{21}{1024}s^{11} + \cdots$$

In general,

$$G_T(s) = \frac{2qs}{1 + \sqrt{1 - 4pqs^2}}$$

$$= qs + pq^2s^3 + 2p^2q^3s^5 + 5p^3q^4s^7 + 14p^4q^5s^9 + 42p^5q^6s^{11} + \cdots$$

Eventual visit to a negative coordinate

The probability of an eventual visit to a negative coordinate is:

$$
\begin{aligned}
\sum_{n\geqslant 0} \mathbb{P}(T=n) &= \sum_{n\geqslant 0} r_n = G_T(1) \\
&= \frac{1-\sqrt{1-4pq}}{2p} = \frac{1-\sqrt{1-4p+4p^2}}{2p} \\
&= \frac{1-\sqrt{(q-p)^2}}{2p} = \frac{1-|q-p|}{2p} \\
&= \begin{cases} 1, & p \leqslant q \\ \dfrac{q}{p}, & p > q \end{cases}
\end{aligned}
$$

Eventual visit to a negative coordinate

Therefore,

- If $p \leqslant q$, the particle visits a negative coordinate with probability 1. In this scenario, T is a proper random variable, and

$$r_n = \mathbb{P}(T = n), \quad n \geqslant 0$$

  represents its probability function.

- If $p > q$, the random variable T is defective. In this case, there is a nonzero probability that the particle never visits a negative coordinate:

$$\mathbb{P}(T = \infty) = 1 - \sum_{n \geqslant 0} \mathbb{P}(T = n) = 1 - \frac{q}{p}$$

Expected time to a negative coordinate

If T is not defective ($p \leqslant q$), then

$$\mathbb{E}(T) = G_T'(1) = \begin{cases} \dfrac{1}{q-p}, & p < q \\ \infty, & p = q \end{cases}$$

For instance,

- for $p = 0.4$ one has $\mathbb{E}(T) = 5$,
- for $p = 0.49$ one has $\mathbb{E}(T) = 50$,
- for $p = 0.499$ one has $\mathbb{E}(T) = 500$ (“long excursions”).

Explicit value of r_n

The probability $r_n = \mathbb{P}(T = n)$ is the coefficient of s^n in the series expansion of

$$G_T(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}$$

Therefore, if c_{2k} is the coefficient of s^{2k} in the expansion of the function

$$\sqrt{1 - 4pqs^2},$$

then, for $k \geqslant 1$,

$$r_{2k-1} = -\frac{1}{2p}\, c_{2k}$$

Explicit value of r_n

To find c_{2k} we can use the binomial series

$$(1+x)^\alpha = \sum_{m=0}^{\infty} \binom{\alpha}{m} x^m,$$

where

$$\binom{\alpha}{m} = \frac{\alpha(\alpha-1)\cdots(\alpha-m+1)}{m!}, \quad \alpha \in \mathbb{R}$$

Thus

$$(1-4pqs^2)^{1/2} = \sum_{k=0}^{\infty} \binom{1/2}{k} (-4pqs^2)^k$$

and

$$c_{2k} = \binom{1/2}{k} (-4pq)^k$$

Explicit value of r_n

Therefore, if $k \geqslant 1$,

$$
\begin{aligned}
r_{2k-1} & \\
&= -\frac{1}{2p}\binom{1/2}{k}(-4pq)^k = -\frac{1}{2}\binom{1/2}{k}(-4)^k p^{k-1}q^k \\
&= -\frac{1}{2}\,\frac{(1/2)(-1/2)(-3/2)\cdots(1/2-k+1)}{k!}\,(-4)^k p^{k-1}q^k \\
&= \frac{1}{2}\,2^k\,\frac{1\cdot 3\cdots(2k-3)}{k!}\,p^{k-1}q^k \\
&= \frac{1\cdot 2\cdot 3\cdots(2k-3)(2k-2)}{(k-1)!\,k!}\,p^{k-1}q^k \\
&= \frac{1}{2k-1}\binom{2k-1}{k}\,p^{k-1}q^k
\end{aligned}
$$

Returns to zero

The probability that the particle is visiting the origin at time $2k$ is

$$\mathbb{P}(X_{2k} = 0) = \binom{2k}{k} p^k q^k$$

- According to Stirling's asymptotic approximation, $m! \sim \sqrt{2\pi m}\,(m/e)^m$, we have

$$\mathbb{P}(X_{2k} = 0) \sim \frac{\sqrt{2\pi 2k}\left(\frac{2k}{e}\right)^{2k}}{2\pi k\left(\frac{k}{e}\right)^{2k}} p^k q^k = \frac{(4pq)^k}{\sqrt{\pi k}}$$

- If the random walk is symmetric, $\mathbb{P}(X_{2k} = 0) \sim \dfrac{1}{\sqrt{\pi k}}$

Returns to zero

Let $A^\star$ be the limit superior of the sequence of events

$$A_k = \{X_{2k} = 0\}\,, \quad k \geqslant 1$$

That is to say, $A^\star$ is the event "the return to 0 happens infinitely often"

Remember that

$$\mathbb{P}(A_k) \sim \frac{(4pq)^k}{\sqrt{\pi k}},$$

so that the series $\sum_{k\geqslant 1} \mathbb{P}(A_k)$ converges if and only if the series $\sum_{k\geqslant 1} (4pq)^k/\sqrt{\pi k}$ converges.

Returns to zero

If the random walk is not symmetric ($p \neq q$), then $4pq < 1$. In this case,

$$\sum_{k=1}^{\infty} \frac{(4pq)^k}{\sqrt{\pi k}} < \infty$$

Therefore, $\sum_k \mathbb{P}(A_k) < \infty$, and the first Borel-Cantelli lemma implies $\mathbb{P}(A^\star) = 0$:

$$\mathbb{P}\left(X_{2k} = 0 \text{ happens infinitely often}\right) = 0$$

If the random walk is not symmetric, then the event $\{X_{2k} = 0\}$ *almost surely occurs a finite number of times.*

First return to zero

Let $f_n = \mathbb{P}(Z = n)$, where $Z = \min\{n \geqslant 1 \colon X_n = 0\}$. For instance,

$$
\begin{aligned}
f_2 &= \mathbb{P}(Z = 2) = 2pq \\
f_3 &= \mathbb{P}(Z = 3) = 0 \\
f_4 &= \mathbb{P}(Z = 4) = 2p^2q^2
\end{aligned}
$$

For $n \geqslant 2$, we have that

$$
\begin{aligned}
f_n &= \mathbb{P}(Z = n) \\
&= p\,\mathbb{P}(Z = n \mid X_1 = 1) + q\,\mathbb{P}(Z = n \mid X_1 = -1) \\
&= p\,r_{n-1} + q\,r'_{n-1}
\end{aligned}
$$

where r'_{n-1} is the probability that the first visit to a positive coordinate happens at time n

First return to zero

Then

$$
\begin{aligned}
G_Z(s) &= p\,s \sum_{n\geqslant 2} r_{n-1} s^{n-1} + q\,s \sum_{n\geqslant 2} r'_{n-1} s^{n-1} \\
&= p\,s \sum_{n\geqslant 0} r_n s^n + q\,s \sum_{n\geqslant 0} r'_n s^n \\
&= p\,s\,G_T(s) + q\,s\,G_{T'}(s) \\
&= p\,s\,\frac{1-\sqrt{1-4pqs^2}}{2ps} + q\,s\,\frac{1-\sqrt{1-4pqs^2}}{2qs} \\
&= 1-\sqrt{1-4pqs^2}
\end{aligned}
$$

First return to zero

The series expansion of $G_Z(s)$ is

$$G_Z(s) = 2pq\, s^2 + 2p^2q^2\, s^4 + 4p^3q^3\, s^6 + 10p^4q^4\, s^8 + \cdots$$

Thus,

$$\begin{aligned}
\mathbb{P}(Z = 2) &= f_2 = 2pq \\
\mathbb{P}(Z = 4) &= f_4 = 2p^2q^2 \\
\mathbb{P}(Z = 6) &= f_6 = 4p^3q^3 \\
\mathbb{P}(Z = 8) &= f_8 = 10p^4q^4 \\
\cdots \quad & \cdots
\end{aligned}$$

Eventual return to zero

The probability of an eventual return to the origin is:

$$\sum_{n \geqslant 0} \mathbb{P}(Z = n) = \sum_{n \geqslant 0} f_n = G_Z(1)$$
$$= 1 - \sqrt{1 - 4pq} = 1 - |q - p|$$

Therefore,

- If $p \neq q$, there is a probability $|q - p| > 0$ that the particle never returns to zero.
- If $p = q$ (symmetric walk), then Z is not defective and the return to zero happens with probability one. However,

$$\mathbb{E}(Z) = G'_Z(1)$$
$$= \frac{d}{ds}\left(1 - \sqrt{1 - s^2}\right)\Big|_{s=1} = \frac{s}{\sqrt{1 - s^2}}\Big|_{s=1} = \infty$$

Other properties

Proposition

The following properties hold:

(a) $\mathbb{P}(Z = 2k) = \dfrac{1}{2k-1}\,\mathbb{P}(X_{2k} = 0)$

(b) *If the random walk is symmetric, then*

$$\mathbb{P}(X_1 \neq 0, X_2 \neq 0, \ldots, X_{2m} \neq 0) = \mathbb{P}(X_{2m} = 0)$$

Other properties

Proof of (a).

$$\mathbb{P}(Z = 2k) = f_{2k} = p\, r_{2k-1} + q\, r'_{2k-1}$$

$$= p\, \frac{1}{2k-1} \binom{2k-1}{k} p^{k-1} q^k + q\, \frac{1}{2k-1} \binom{2k-1}{k} p^k q^{k-1}$$

$$= \frac{1}{2k-1} \left(\binom{2k-1}{k} + \binom{2k-1}{k-1} \right) p^k q^k$$

$$= \frac{1}{2k-1} \binom{2k}{k} p^k q^k = \frac{1}{2k-1}\, \mathbb{P}(X_{2k} = 0)$$

Other properties

Proof of (b).

Let $f_{2k} = \mathbb{P}(Z = 2k)$ and $u_{2k} = \mathbb{P}(X_{2k} = 0)$. If $p = q$, then

$$u_{2k} = \binom{2k}{k}\frac{1}{2^{2k}}, \quad f_{2k} = \frac{u_{2k}}{2k-1} = \frac{1}{2k-1}\binom{2k}{k}\frac{1}{2^{2k}},$$

and it is easily checked that $u_{2k-2} - u_{2k} = f_{2k}$, for all $k \geqslant 1$.

Hence,

$$\begin{aligned}\mathbb{P}(X_1 \neq 0, \ldots, X_{2m} \neq 0) &= \mathbb{P}(Z > 2m) = 1 - \mathbb{P}(Z \leqslant 2m) \\ &= 1 - \sum_{k=1}^{m} f_{2k} = 1 - \sum_{k=1}^{m}(u_{2k-2} - u_{2k}) = u_{2m} \\ &= \mathbb{P}(X_{2m} = 0)\end{aligned}$$

Last equalization

Proposition

The probability $\alpha_{2k,2m}$ that a symmetric random walk of length $2m$ has a last return to zero at time $2k$, $0 \leqslant k \leqslant m$, equals

$$\alpha_{2k,2m} = \binom{2k}{k}\binom{2m-2k}{m-k}\frac{1}{2^{2m}}$$

$$= \mathbb{P}(X_{2k} = 0)\,\mathbb{P}(X_{2m-2k} = 0)$$

- Notice that $k = 0$ corresponds to $\{X_n \neq 0 \colon 1 \leqslant n \leqslant 2m\}$, and $k = m$ corresponds to $\{X_{2m} = 0\}$.

Last return equalization

Proof.

Let $A_{2k,2m}$ denote the event that the symmetric random walk of length $2m$ has its last return to zero at time $2k$, where $0 \leqslant k \leqslant m$.

Then,

$$
\begin{aligned}
\alpha_{2k,2m} &= \mathbb{P}(A_{2k,2m}) \\
&= \mathbb{P}\left(\{X_{2k} = 0\} \cap \{X_{2k+1} \neq 0, X_{2k+2} \neq 0, \ldots, X_{2m} \neq 0\}\right) \\
&= \mathbb{P}(X_{2k} = 0)\, \mathbb{P}(X_{2k+1} \neq 0, X_{2k+2} \neq 0, \ldots, X_{2m} \neq 0 \,|\, X_{2k} = 0)
\end{aligned}
$$

Last equalization

Moreover, since the random walk is temporally homogeneous,

$$
\begin{aligned}
\mathbb{P}(X_{2k+1} \neq 0, X_{2k+2} \neq 0, \ldots, X_{2m} \neq 0 \,|\, X_{2k} = 0)& \\
&= \mathbb{P}(X_1 \neq 0, X_2 \neq 0, \ldots, X_{2m-2k} \neq 0 \,|\, X_0 = 0) \\
&= \mathbb{P}(X_0 = 0, X_1 \neq 0, X_2 \neq 0, \ldots, X_{2m-2k} \neq 0) \\
&= \mathbb{P}(X_1 \neq 0, X_2 \neq 0, \ldots, X_{2m-2k} \neq 0)
\end{aligned}
$$

Hence,

$$
\begin{aligned}
\alpha_{2k,2m} &= \mathbb{P}(X_{2k} = 0)\,\mathbb{P}(X_1 \neq 0, X_2 \neq 0, \ldots, X_{2m-2k} \neq 0) \\
&= \mathbb{P}(X_{2k} = 0)\,\mathbb{P}(X_{2m-2k} = 0)
\end{aligned}
$$

Last equalization

For large k, the asymptotic behaviour of $\mathbb{P}(X_{2k} = 0)$ is

$$\mathbb{P}(X_{2k} = 0) \sim \frac{1}{\sqrt{\pi k}}$$

Hence, if k and m go to infinity, so that $m - k$ s large, then

$$\alpha_{2k,2m} \sim \frac{1}{\pi \sqrt{k(m-k)}}$$

Last equalization

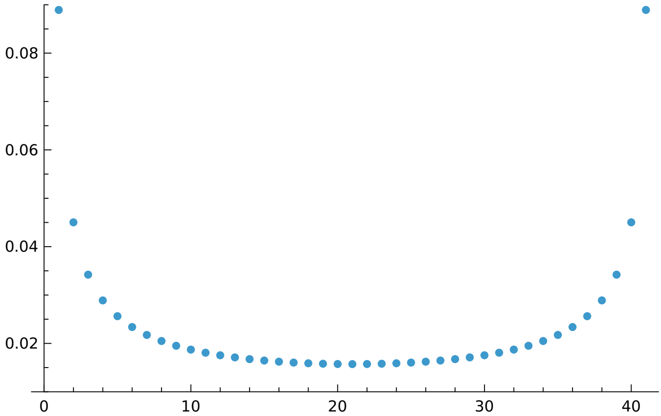


Figure: A plot of the probability $\alpha_{2k,80}$ for $k = 0, 1, \ldots, 40$

Last equalization

Observe that $\alpha_{2k,2m} = \alpha_{2m-2k,2m}$.

Given that each possible trajectory of length $2m$ occurs with equal probability, the symmetry $\alpha_{2k,2m} = \alpha_{2m-2k,2m}$ indicates that the number of paths returning to zero for the last time at time $2k$ equals the number of paths returning to zero for the last time at time $2m - 2k$.

Therefore, the probability that a symmetric random walk of length $2m$ does not return to zero during the final m steps is $1/2$.

Last equalization

Another surprising result is the following:

Proposition

If players A and B play a game of length $2m$, the probability that A will be in the lead exactly $2k$ times is equal to $\alpha_{2k,2m}$.

While it is commonly assumed that the most probable number of times player A leads is m, this assumption does not hold.

Contrary to intuition, the least probable number of times player A leads occurs when $2k = m$, while the most probable values are observed at $2k = 0$ and $2k = 2m$.